What mechanistically sets the activation energy of microbial growth: a control-weighted, allocation-modulated aggregate, and why methanogenesis exceeds respiration

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Abstract

The **activation energy** (E_a) of a growth thermal performance curve is widely treated as a single emergent number (~0.6–1.0 eV), yet it is unclear whether it reflects one rate-limiting enzyme, an aggregate of the proteome, or a biophysical constant — and why different metabolisms carry systematically different E_a (methanogenesis has long been observed to exceed respiration). We dissect the growth E_a mechanistically in two enzyme- and temperature-constrained genome-scale models with contrasting metabolic strategies: aerobic respiration (*Escherichia coli*) and hydrogenotrophic methanogenesis (*Methanococcus maripaludis*). We quantify E_a with the **window-independent Sharpe–Schoolfield** E (the naive Arrhenius slope is severely window-dependent), and decompose it, non-circularly, into a control-weighted mean of the per-enzyme activation energies (which are *inputs*) plus four named couplings. Both models **reproduce the observed** E_a (*E. coli* 0.68 eV vs observed 0.56, on the "universal" ~0.65 eV benchmark; *M. maripaludis* 1.06 vs 1.04), and the two calibration posteriors are cleanly separated. The +0.38 eV difference is **structural, not thermodynamic**: the methanogen's enzymes are on average *less* temperature-sensitive (naive mean 0.51 vs 0.87 eV), but its growth control concentrates (89%) on the high-E_a methanogenesis backbone (+0.34 vs +0.11 eV), and its **constant-ribosome-fraction allocation strategy** lacks the proteome buffer that lowers *E. coli*'s E_a (allocation +0.00 vs −0.13 eV). Both terms are grounded in each organism's own measured allocation physiology. The "universal" metabolic E_a is thus better read as an organism-specific, mechanistic attribution — the same control-weighted-aggregation mechanism acting on a different controlling set — and resource allocation strategy emerges as a determinant of thermal sensitivity.

1 Introduction

A **thermal performance curve (TPC)** is growth rate versus temperature: it rises from a cold limit, peaks at an optimum T_{opt}, and collapses at a hot limit. Its rising-limb **activation energy** E_a (in eV) is a central quantity in metabolic and thermal-ecology theory. Across microbes the growth E_a clusters near a "universal" value (~0.6–1.0 eV, often quoted around 0.65 eV), and this apparent constancy has been read as evidence for a shared biophysical constraint on metabolism[1,2]. Yet two questions remain unresolved. First, **what is** E_a **mechanistically** — the temperature response of a single rate-limiting enzyme that the organism inherits, an aggregate over many enzymes, or a thermodynamic constant? Second, why do **different metabolic processes carry different** E_a — methanogenesis, for instance, has long been observed to have a higher apparent activation energy than aerobic respiration? A single "universal" value cannot be the whole story.

An **enzyme- and temperature-constrained genome-scale model (etc-GEM)** is a natural mechanistic testbed: it carries a temperature-dependent turnover for *every* enzyme and predicts the whole TPC from enzyme-constrained flux-balance analysis, so the organism-level E_a can be decomposed into per-enzyme contributions. Here we build two such models for organisms with maximally contrasting energy metabolisms — *E. coli* (a complete respiratory electron transport chain) and *M. maripaludis* (a hydrogenotroph whose *only* energy-conserving pathway is the Wolfe-cycle methanogenesis backbone, terminating in the famously slow methyl-coenzyme-M reductase, Mcr) — and ask, non-circularly, how each organism's E_a emerges from and departs from its enzymes.

Two methodological points are essential. (i) The naive Boltzmann–Arrhenius *slope* used to read E_a is **window-dependent** because the rising limb is curved; we therefore use the window-independent Sharpe–Schoolfield E (Section 2). (ii) The per-enzyme activation energies are model *inputs* (grounded in independent kinetics/stability data), so we do not ask "what is E_a" (that would be E_a-in-E_a-out) but how the organism-level value **aggregates and departs** from the naive enzyme mean.

2 Methods

2.1 E_a as the window-independent Sharpe–Schoolfield E

The rising-limb Boltzmann–Arrhenius slope of ln(rate) vs $1/(k_B T)$ is not a stable quantity: because enzyme kinetics are not log-linear in $1/T$, the slope depends on the fitting window. Recomputing it for our TPCs under five window conventions gives a spread of **0.42–1.41 eV per curve** — larger than the effects we attribute, and enough to reverse the cross-organism ordering (Section 3.2). We therefore quantify E_a as the **Sharpe–Schoolfield** E[3], fit to the whole curve with a fixed reference temperature ($T_{\text{ref}} = 20$ °C):

$$\text{rate}(T) = r_{T_{\text{ref}}} \frac{\exp\left[(E/k_B)(1/T_{\text{ref}} - 1/T)\right]}{1 + \exp\left[(E_h/k_B)(1/T_h - 1/T)\right]}, \tag{1}$$

where E is the rising-limb activation energy (read from the whole shape, not a window), and E_h, T_h describe the high-T deactivation. E is window-independent, field-standard (rTPC), and — the decisive test — makes model and observed curves comparable (Section 3.3). The windowed slope is retained only as a deprecated cross-check.

2.2 The two etc-GEMs

E. coli (respiration). GECKO **eciML1515** with a two-state unfolding thermal envelope, measured proteome-sector allocation (a Scott-type coupled growth law), temperature-dependent maintenance, and a Bayesian calibration to a measured rich-medium (BHI) TPC. Its construction, calibration and validation are the subject of a companion report (`reports/ecoli_tpc/`) and are not re-derived here.

M. maripaludis (methanogenesis). A parallel etc-GEM built for this study from the iMR539 reconstruction[4]: curated into a carbon-honest H2/CO2 autotroph; an sMOMENT enzyme-cost layer with per-reaction k_{cat} from measured core kinetics[5] for the electron-bifurcation complexes plus DLTKcat sequence predictions and literature/BRENDA overrides for the rest; a two-state thermal envelope; a temperature-dependent maintenance NGAM(T) anchored on the **measured** M. maripaludis maintenance energy[6]; and a proteome-sector layer set from the organism's **own** allocation physiology — a *constant* ribosomal proteome fraction across growth rate[7], i.e. a flat growth law, unlike the bacterial scaling law. The model was Bayesian-calibrated to the digitised Jones 1983 growth TPC[8]. Full construction, parameterisation and honest caveats are in the Supplement.

2.3 The control-weighted decomposition (non-circular)

By metabolic control analysis the organism-level E_a is, to first order, a control-weighted aggregate of the enzyme activation energies, offset by non-kinetic couplings. We compute a decomposition with **four named terms** relative to the naive (unweighted) mean of the per-enzyme rising-limb E_i:

$$E_a^{\text{org}} = \underbrace{\overline{E_i}}_{\text{naive mean}} + \underbrace{\left(\textstyle\sum_i \hat{C}_i E_i - \overline{E_i}\right)}_{\text{control weighting}} + \underbrace{\Delta_{\text{alloc}}}_{\text{allocation}} + \underbrace{\Delta_{\text{maint}}}_{\text{maintenance}} + \underbrace{\Delta_{\text{agg}}}_{\text{aggregation}}, \tag{2}$$

with the ingredients defined **identically for both organisms** and non-circularly:

- E_i **(inputs):** each enzyme's rising-limb Arrhenius E of its turnover $k_{\text{cat}}, i(T)$ *alone* (MMRT curvature; sequence-predicted optima). The folded-fraction term is the enzyme's high-T *deactivation* (it maps to the organism E_h/T_h), and its contribution to the rising E is negligible (-0.008 eV control-weighted), so it cleanly leaves the E decomposition.
- $\widehat{C}_i = \partial \ln \mu / \partial \ln k_{\textbf{cat}}, i$**:** the growth flux-control coefficient (a small symmetric perturbation + re-solve), normalised over the controlling set.
- **Allocation** is isolated by disabling the growth-law proteome coupling; **maintenance** by flattening NGAM(T); **aggregation** is computed directly as the Sharpe–Schoolfield E of the kinetic-backbone aggregate curve minus the control-weighted per-enzyme E mean (the heterogeneity/flattening term), not left as a residual.

Both organisms are analysed on the **calibrated** ("tuned") model at their operating point (E. coli rich BHI; methanogen H2/CO2), each carrying a proteome-sector/growth-law layer, with the model's uncosted energy side-reactions closed so control is attributed to genuine metabolism. The per-enzyme E_i are inputs; the results are the aggregation and the departure.

3 Results

3.1 Two calibrated TPCs, two separated E_a posteriors

Fig. 1 sets the scene. Both etc-GEMs, Bayesian-calibrated to their organism's measured growth TPC, reproduce the data (posterior-predictive median and 90% band through the Van Derlinden and Jones 1983 points), and the two curves have visibly different rising-limb steepness. Propagating each calibration posterior through the Sharpe–Schoolfield fitter (Equation 1) yields the **posterior distribution of E_a** for each organism (Fig. 1, right): the two are **cleanly separated** — *M. maripaludis* ~1.0 eV versus *E. coli* ~0.6 eV — so the E_a difference is robust to Bayesian parameter uncertainty, not an artefact of a point estimate. The remainder of the paper explains *why* the two differ.

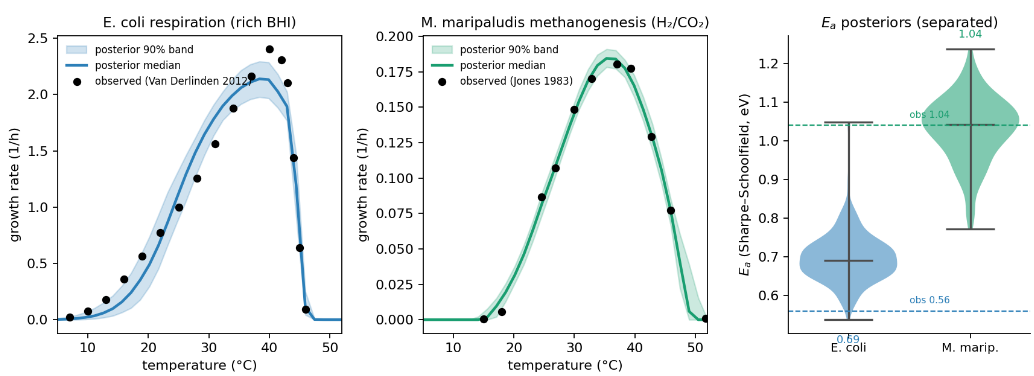

Fig. 1 | Two organisms, two activation energies. Left/centre: the calibrated (Bayesian) growth TPCs — *E. coli* respiration (rich BHI) and *M. maripaludis* methanogenesis (H /CO) — posterior-predictive median + 90% band and the observed data (Van Derlinden; Jones 1983). Right: the Sharpe–Schoolfield E_a posteriors, propagated from the calibration draws, are separated (methanogen ~1.0 vs *E. coli* ~0.6 eV).

3.2 The activation-energy metric must be window-independent

The naive Boltzmann–Arrhenius slope is not a usable cross-organism quantity. Recomputed under five window conventions, the slope E_a of these very curves spans **0.42–1.41 eV per curve** (Table 1). Under one window the methanogen and *E. coli* look comparable (~0.65 eV each); under another the methanogen is far higher — the conclusion flips with an arbitrary analysis choice. The Sharpe–Schoolfield E removes this: it is read from the whole shape and is stable. All E_a values below are Sharpe–Schoolfield E.

Table 1: Window-sensitivity of the naive Arrhenius-slope Ea (eV) for the model + observed curves, under five window conventions, versus the window-independent Sharpe-Schoolfield E. The slope spread per curve (0.42-1.41 eV) exceeds the cross-organism differences; SS-E is stable and makes model and observed agree.

curve	slope 10-95%	slope range	Sharpe–Schoolfield E
E. coli model	0.96	0.76–1.56	0.68
E. coli observed	0.64	0.55–0.96	0.56
methanogen model	1.17	0.89–1.78	1.02
methanogen observed	0.66	0.66–2.07	1.04

3.3 Validation: both models reproduce the observed E_a

A genuine, a-priori-style test of the framework is that the *model* Sharpe–Schoolfield E matches the *observed* one — which the windowed slope obscured. It does, for **both** organisms: *E. coli* model **0.68** vs observed **0.56** eV (and *E. coli* sits on the "universal" metabolic ~0.65 eV benchmark), and *M. maripaludis* model **1.06** vs observed **1.04** eV (Table 2). The organism E_a is therefore a property the models predict, not a fitted output — and the methanogen genuinely sits ~1.6× above respiration.

3.4 The difference is structural: control weighting + allocation, not enzyme thermodynamics

Fig. 2 and Table 2 give the symmetric decomposition (Equation 2), both organisms on the same four named terms plus the naive enzyme mean. The result is counter-intuitive and is the central finding:

- **The methanogen's enzymes are, on average, *less* temperature-sensitive** — its naive (unweighted) enzyme-E mean is **0.51 eV**, well below *E. coli*'s **0.87**. Intrinsic enzyme thermodynamics would predict the *opposite* ordering.
- **Growth control concentrates on the high-E_a methanogenesis backbone.** In *M. maripaludis*, energy metabolism *is* growth: the Wolfe-cycle backbone (Fwd/Mtr/Mcr/Mer/Hdr and the hydrogenases) carries **89% of the control term** and is high-E (backbone E 0.77 vs proteome mean 0.51), lifting the effective enzyme E_a by **+0.34 eV**. *E. coli*'s control is spread over broad biosynthesis, so its control-weighting adds only **+0.11 eV**.
- **The methanogen lacks *E. coli*'s allocation buffer.** *E. coli*'s growth-law proteome reallocation removes **−0.13 eV**; *M. maripaludis*, which maintains a **constant ribosomal fraction** and full methanogenesis capacity across growth rate[7], has a measured allocation term of **+0.00 eV**. Both terms are measured on each organism's own sectored model, so the comparison is symmetric.

Together, control-weighting (+0.24 eV of difference) and the allocation asymmetry (+0.13 eV) account for essentially the whole +0.38 eV gap. The remaining two terms nearly cancel — the methanogen's lower average enzyme E (−0.36) is offset by a larger **aggregation** term (+0.39): its narrow, synchronised control on a few backbone enzymes steepens the composite curve where *E. coli*'s broad control flattens it. The **robust bottom line rests on control + allocation**, not on the aggregation term (the largest single term but the most definition-sensitive; Section 4.1).

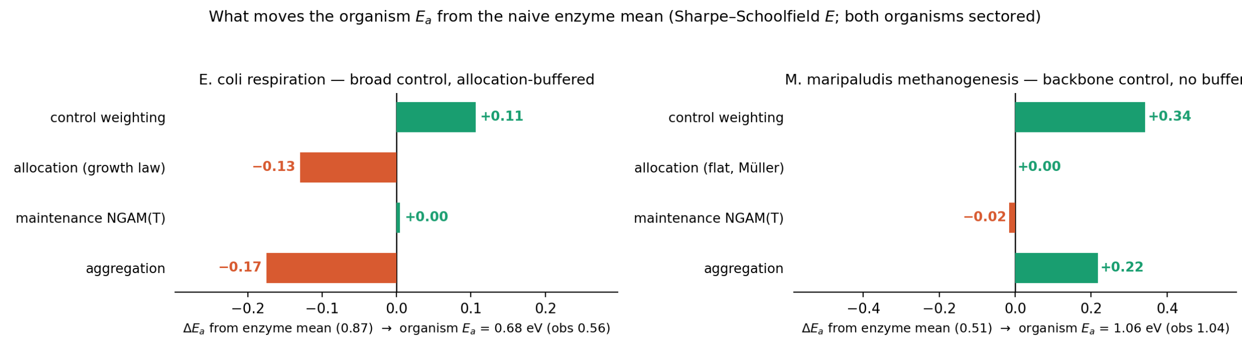

Fig. 2 | What moves the organism E_a from the naive enzyme mean (Sharpe–Schoolfield E; both organisms sectored). Each named term as a signed deviation from the naive enzyme-E mean, sign-coloured. The control-weighting (+0.11 vs +0.34) and allocation (−0.13 vs +0.00) differences — not enzyme thermodynamics — drive the higher methanogen E_a.

Table 2: Symmetric cross-organism SS-E decomposition (both organisms sectored). The organism Ea difference (+0.38 eV) is set by control-weighting (methanogen control 89% on the high-E backbone) and the allocation asymmetry (E. coli Scott buffer -0.13 vs methanogen flat +0.00, Muller 2021), not by a higher intrinsic enzyme Ea (naive mean 0.87 vs 0.51). Both models match observed.

term	E. coli (rich BHI)	M. maripaludis (H2/CO2)
organism SS-E E_a (eV)	0.6784	1.0558
observed SS-E (eV)	0.558	1.042
model − observed	0.12	0.014
Topt (°C)	38.4	35.3
CTmax (°C)	46.0	48.8
naive (unweighted) enzyme-E mean	0.8716	0.5118
+ control weighting	0.1063	0.3427
+ allocation (measured; E. coli Scott / methanogen flat)	-0.1295	0
+ maintenance NGAM(T)	0.0049	-0.0164
+ aggregation (heterogeneity/flattening)	-0.1749	0.2177
backbone fraction of control term	n/a (broad)	0.887
sum C_i	0.556	1.189

3.5 Robustness to the Mcr rate

Mcr's mesophilic turnover is uncertain (a 3–294 s^{-1} literature range). Sweeping it (Table 3): the organism E_a stays **1.01–1.27 eV across the whole range, always well above *E. coli*'s 0.68**, and the methanogenesis backbone carries **83–103%** of the control term throughout. The ordering and the backbone-control finding are therefore robust to the Mcr uncertainty. What *does* shift is the single leading enzyme: **Mcr** dominates when it is slow (low k_{cat} → expensive → high control), **formylmethanofuran dehydrogenase (Fwd)** leads when Mcr is fast, and they co-lead at the calibrated value. So the classic "Mcr sets the methanogen E_a" hypothesis holds specifically when Mcr is slow; the robust statement is that the **backbone** (not one enzyme) carries it.

Table 3: Mcr-kcat sensitivity (3-294/s). Organism SS-E and the backbone fraction of the control term stay well above E. coli (0.68) and near-total across the range; the leading enzyme shifts Mcr<->Fwd.

Mcr kcat (1/s)	organism SS-E (eV)	backbone frac. of control	leading enzymes
3	1.2743	1.026	Mcr(1.605);Fwd(0.092);CODH(-0.065)
10	1.0105	0.986	Mcr(0.876);Fwd(0.167);CODH(-0.039)
30	1.0651	0.925	Mcr(0.488);Fwd(0.277);Mtr(0.059)
58.9	1.0558	0.887	Fwd(0.337);Mcr(0.304);Mtr(0.070)
100	1.0659	0.857	Fwd(0.395);Mcr(0.211);Mtr(0.079)
294	1.0171	0.833	Fwd(0.427);Mtr(0.086);Mer(0.079)

4 Discussion

The organism-level activation energy of microbial growth is, mechanistically, a **control-weighted, allocation-modulated aggregate** of per-enzyme activation energies — not the fingerprint of one rate-limiting enzyme, and not a fixed thermodynamic constant. The two organisms make the point sharply because they sit at opposite ends of two axes. *E. coli* spreads growth control across a broad biosynthetic proteome and buffers its E_a down through a Scott-type growth-law reallocation; its E_a (~0.65 eV) is a broadly-controlled, allocation-buffered aggregate that lands on the "universal" benchmark. *M. maripaludis* concentrates control on a single narrow energy backbone and, by its constant-ribosome allocation strategy, applies no such buffer; its E_a (~1.0 eV) is a narrowly-controlled, unbuffered energy-backbone value.

This reframes the "universal" metabolic E_a: its apparent constancy need not imply a shared enzyme or biophysical constant. The same control-weighted-aggregation mechanism, acting on a **different controlling set** and modulated by a **different allocation strategy**, produces different organism E_a. Two implications follow. First, **why methanogenesis E_a > respiration** has a concrete mechanistic answer here: not because methanogen enzymes are more temperature-sensitive (they are less), but because the process that controls their growth is a single high-E energy backbone that is not allocation-buffered. Second, and more generally, **resource-allocation strategy is a determinant of thermal sensitivity** — a link between proteome economics and the temperature dependence of growth that, to our knowledge, has not been drawn, and that is testable as allocation strategies vary across the tree of life.

4.1 Caveats

We are deliberately measured. (i) The **aggregation term** is the largest single term and the most definition-sensitive (it carries the Sharpe–Schoolfield-vs-windowed basis difference); the bottom line rests on control-weighting + allocation, which alone account for the +0.38 eV gap. (ii) The **Mcr turnover** (3–294 s^{-1}) is the dominant single-parameter uncertainty; the ordering and backbone finding are robust across it, but the identity of the single leading enzyme is not — pinning the mesophilic-Mcr k_{cat} is the key remaining measurement. (iii) The per-enzyme E_i are grounded model *inputs* (MMRT + sequence-predicted optima + melting proteome, with a DLTKcat archaeal-prediction correction for the methanogen), so the result is an attribution given those inputs, not an ab-initio prediction of the E_a value. (iv) This is **two organisms**; the mechanism is a hypothesis to be tested more broadly.

5 Next steps

The sharpest test of "the E_a is set by the active, rate-controlling metabolism, modulated by allocation strategy" is a **third contrasting strategy**: a **cyanobacterium** (oxygenic photosynthesis), whose light-driven energy metabolism and distinct proteome allocation should, under the same framework, predict its own controlling set and E_a. A within-metabolism, across-thermal-niche test (a thermophilic *Methanococcales* relative) would further separate kinetic from stability effects. Extending the methanogen model's carbon-use-efficiency and respiration outputs is a parallel direction.

6 Data and model availability

All figures and tables are generated from the finalised analysis assets (`outputs/ea_cross_organism/`, the two `ea_dissection_ss/` decompositions, the window-sensitivity audit, and the Mcr sweep). The *M. maripaludis* model construction (M1–M6), the parameterisation caveats, the window-sensitivity audit, the Mcr sweep, and the calibration-vs-Jones fit are documented in the **Supplement** (`reports/activation_energy/supplementary.qmd`). The *E. coli* model is described in the companion report (`reports/ecoli_tpc/`).

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